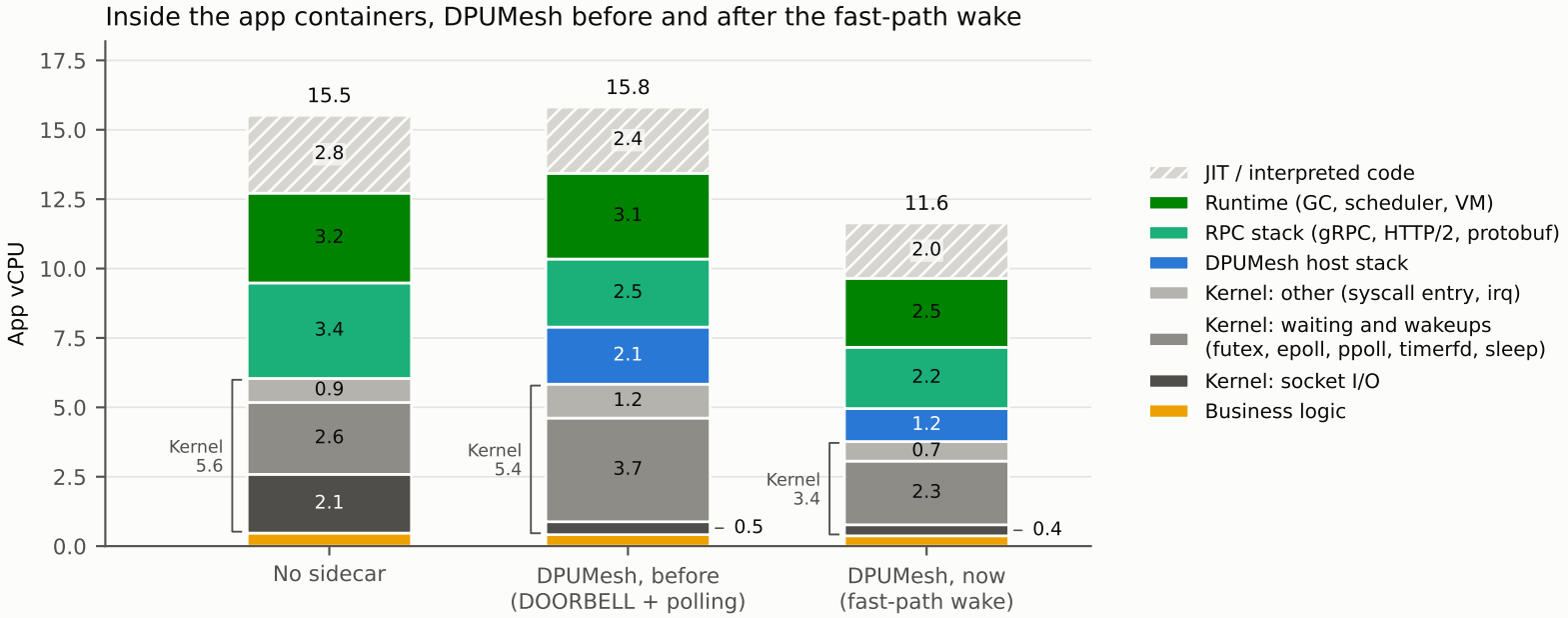
Inside the app containers, DPUMesh before and after the fast-path wake
App vCPU
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
15.5
2.8
3.2
3.4
0.9
2.6
2.1
Kernel
5.6
15.8
2.4
3.1
2.5
2.1
1.2
3.7
Kernel
5.4
– 0.5
11.6
2.0
2.5
2.2
1.2
0.7
2.3
Kernel
3.4
– 0.4
No sidecar
DPUMesh, before
(DOORBELL + polling)
DPUMesh, now
(fast-path wake)
JIT / interpreted code
Runtime (GC, scheduler, VM)
RPC stack (gRPC, HTTP/2, protobuf)
DPUMesh host stack
Kernel: other (syscall entry, irq)
Kernel: waiting and wakeups
(futex, epoll, ppoll, timerfd, sleep)
Kernel: socket I/O
Business logic